1. $\int \frac{1}{\sin x}\,dx$

$\int \frac{1}{\sin x}\,dx$

$= \int \frac{\sin x}{\sin^2 x}$

$= \int \frac{\sin x}{(1-\cos^2 x)}\,dx$

$= -\int \frac{d\cos x}{1-\cos^2 x}$

$= -\frac{1}{2}\left(\int \frac{d\cos x}{1+\cos x} + \frac{d\cos x}{1-\cos x}\right)$

$= -\frac{1}{2}\ln(1+\cos x) + \frac{1}{2}\ln(1-\cos x) + C$

2. ★

$\int \sqrt{x^2\pm a^2}\,dx = \frac{x}{2}\sqrt{x^2\pm a^2} \pm \frac{a^2}{2}\ln|x+\sqrt{x^2\pm a^2}| + C$

$\int \sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\arcsin\frac{x}{a} + C$

$\int \frac{dx}{\sqrt{x^2\pm a^2}} = \ln|x+\sqrt{x^2\pm a^2}| + C$

$\int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin\frac{x}{a} + C$

3. ★ $\int \frac{x e^{\arctan x}}{(1+x^2)^{\frac{3}{2}}}\,dx$

$= \int \frac{x}{\sqrt{1+x^2}}\,d e^{\arctan x}$

$= \frac{x}{\sqrt{1+x^2}} e^{\arctan x} - \int e^{\arctan x}\frac{1}{(1+x^2)^{\frac{3}{2}}}\,dx$

$= \frac{x e^{\arctan x}}{\sqrt{1+x^2}} - \int \frac{1}{\sqrt{1+x^2}}\,d e^{\arctan x}$

$= \frac{x e^{\arctan x}}{\sqrt{1+x^2}} - \frac{e^{\arctan x}}{\sqrt{1+x^2}} - \int \frac{x e^{\arctan x}}{(1+x^2)^{\frac{3}{2}}}\,dx$

$\Rightarrow \frac{1}{2}\frac{x e^{\arctan x}}{\sqrt{1+x^2}} - \frac{1}{2}\frac{e^{\arctan x}}{\sqrt{1+x^2}} + C$

4. $\int \frac{dx}{\sqrt{1+e^x}}$

令 $\sqrt{1+e^x} = t$

$e^x = t^2-1 \quad x = \ln(t^2-1)$

$\frac{dx}{dt} = \frac{2t}{t^2-1}$

$\int \frac{1}{t}\cdot\frac{2t}{t^2-1}\,dt$ 略

5. $\int \frac{\arctan\sqrt{x}}{\sqrt{x}\,(1+x)}\,dx$

注意到

$\int 1\, d(\arctan\sqrt{x})^2$

$= (\arctan\sqrt{x})^2 + C$

6. $\int \frac{x^5}{x^6-x^3-2}\,dx$

$= \int \frac{x^3\cdot x^2}{(x^3-2)(x^3+1)}\,dx$

$= \int \frac{x^3}{(x^3-2)(x^3+1)}\cdot\frac{dx^3}{3}$

略

7. ★ $\int \cancel{\sin}\ \int \sqrt{x^2-a^2}\,dx$

$= x\cdot\sqrt{x^2-a^2} - \int x\cdot\frac{x}{\sqrt{x^2-a^2}}\,dx$

$= x\sqrt{x^2-a^2} - \int \frac{x^2-a^2+a^2}{\sqrt{x^2-a^2}}\,dx$

$= x\sqrt{x^2-a^2} - \int\sqrt{x^2-a^2}\,dx + a^2\int\frac{1}{\sqrt{x^2-a^2}}\,dx$

$\int \sqrt{x^2-a^2}\,dx = \frac{x\sqrt{x^2-a^2}}{2} + \frac{a^2}{2}\ln|x+\sqrt{x^2-a^2}| + C$

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(1) $\int \frac{\arctan x}{x^2(1+x^2)}dx$

$= \int \frac{\arctan x}{x^2}dx - \int \frac{\arctan x}{1+x^2}dx$

$= -\int \arctan x \, d\frac{1}{x} - \frac{1}{2}(\arctan x)^2 + C$

$= -\frac{\arctan x}{x} + \int \frac{1}{x(1+x^2)}dx - \frac{(\arctan x)^2}{2} + C$

$\int \frac{1}{x(1+x^2)}dx$

$= \int \left(\frac{-x}{1+x^2} + \frac{1}{x}\right)dx$

$= -\frac{\ln(1+x^2)}{2} + \ln x$

5 略

(2) $\int \frac{dx}{\cos^3 x}$

$= \int \frac{\sin^2 x + \cos^2 x}{\cos^3 x}dx$

$= \int (\tan^2 x \sec x + \sec x)dx$

$= \int \tan^2 x \sec x \, dx + \int \sec x \, dx$

$= \int \tan x \, d\sec x + \int \sec x \, dx$

$= \tan x \sec x - \int \sec^3 x \, dx + \int \sec x \, dx$

$\frac{\tan x \sec x + \ln|\sec x + \tan x|}{2} + C$

(3) $\int \arccos x \, dx$

$= x\arccos x - \cancel{\int x \cdot \frac{1}{1+\cos^2 x}}$

$= x\arccos x - \int -\frac{x}{\sqrt{1-x^2}}dx$

$= x\arccos x + \int \frac{x}{\sqrt{1-x^2}}dx$

$= x\arccos x - \sqrt{1-x^2} + C$

(4) $\int \frac{1}{\sin x + \tan x}dx$

$= \int \frac{\cos x}{\sin x\cos x + \sin x}dx$

$= \int \frac{\cos x}{\sin x(1+\cos x)}dx$

$= -\int \frac{\cos x \, d\cos x}{(1-\cos^2 x)(1+\cos x)}$

略

(5) $\cancel{\int x \arctan^2 x \, dx}$

(5) $\int \sin(\ln x)dx$

$= x\sin(\ln x) - \int x \cdot \cos(\ln x) \cdot \frac{1}{x}dx$

$= x\sin(\ln x) - x\cos(\ln x) + \int x\sin(\ln x) \cdot \frac{-1}{x}dx$

$\frac{x\sin(\ln x) - x\cos(\ln x)}{2}$

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系别________ 班级________ 姓名________ 第 页

(1) $\int \frac{x^3}{\sqrt{1+x^2}} dx$ 或

$= \int \frac{x^2 \cdot x}{\sqrt{1+x^2}} dx$ $\quad \int \frac{x}{\sqrt{1+x^2}} \cdot x^2 dx$

$\quad = \int x^2 d\sqrt{1+x^2}$

$= \frac{1}{2}\int \frac{x^2}{\sqrt{1+x^2}} dx^2$

$= \frac{1}{2}\int \frac{t}{\sqrt{1+t}} dt$

$\sqrt{1+t} = m \quad t = m^2-1$

$\frac{dt}{dm} = 2m$

$= \frac{1}{2}\int \frac{m^2-1}{\cancel{m}} \cdot 2\cancel{m} dm$

$= \int (m^2-1) dm$ 略

(2) $\int \frac{x+1}{x^2+x+1} dx$

$= \frac{1}{2}\int \frac{2x+1}{x^2+x+1} dx + \frac{1}{2}\int \frac{1}{x^2+x+1} dx$

化简易错

$= \frac{1}{2}\ln(x^2+x+1) + \frac{1}{2}\int \frac{1}{(x+\frac{1}{2})^2+\frac{3}{4}} dx$

$= \frac{1}{2}\ln|x^2+x+1| + \frac{1}{\sqrt{3}}\arctan\left(\frac{2}{\sqrt{3}}(x+\frac{1}{2})\right) + C$

(3) $\int \sec^8 x dx$

$= \int \sec^6 x \, d\tan x$

$= \int (\tan^2 x+1)^3 d\tan x$

$\cancel{\int (t^2+1)^3 dt}$

$= \cancel{\ldots}$

化开即可

(4) $\int \frac{1-\ln x}{(x+\ln x)^2} dx$

$= \int \frac{1-\ln x}{x^2(1+\frac{\ln x}{x})^2} dx$

$= \int \frac{1}{(1+\frac{\ln x}{x})^2} d\frac{\ln x}{x}$

略

(5) $\int \frac{x+1}{x(1+xe^x)} dx$

$= \int \frac{e^x(x+1)}{xe^x(1+xe^x)} dx$

$= \int \frac{1}{xe^x(1+xe^x)} d\,xe^x$

略

(6) $\int \frac{dx}{1+a\cos x} dx$

中

$= \cancel{\int}$ 令 $t = \tan\frac{x}{2}$

$\int \frac{2dx}{t^2(1-a)+1+a} dt$

$0<a<1$

$a \ngtr 1$ 时

$\frac{2}{1-a}\int \frac{1}{t^2+(1+a)/(1-a)} dt$

$= \frac{2}{1-a} \frac{2}{\sqrt{1-a^2}} \arctan\left(\sqrt{\frac{1-a}{1+a}} \tan\frac{x}{2}\right) + C$

$a \geq 1$ 时

$\frac{2}{a-1}\int \frac{1}{\frac{a+1}{a-1} - t^2} dt$

$= \frac{2}{a-1} \cdot \frac{1}{2\sqrt{\frac{a+1}{a-1}}} \ln \frac{\sqrt{\frac{a+1}{a-1}} + t}{\sqrt{\frac{a+1}{a-1}} + t} + C$

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(1) $\int \frac{1}{1+x^4}dx$

$= \frac{1}{2}\int\left(\frac{x^2+1}{1+x^4} - \frac{x^2-1}{x^4+1}\right)dx$

$= \frac{1}{2}\int \frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}} - \frac{1}{2}\int\frac{1-\frac{1}{x^2}}{x^2+\frac{1}{x^2}}dx$

$= \frac{1}{2}\int\frac{d(x-\frac{1}{x})}{(x-\frac{1}{x})^2+2} - \frac{1}{2}\int\frac{d(x+\frac{1}{x})}{(x+\frac{1}{x})^2+2-2}$

$= \frac{1}{2}\cdot\frac{1}{\sqrt{2}}\arctan\left(\frac{x-\frac{1}{x}}{\sqrt{2}}\right)$

$\quad - \frac{1}{2}\cdot\frac{1}{2\sqrt{2}}\ln\left|\frac{x+\frac{1}{x}-\sqrt{2}}{x+\frac{1}{x}+\sqrt{2}}\right| + C$

$= \frac{1}{2\sqrt{2}}\arctan\frac{x^2-1}{\sqrt{2}x} - \frac{1}{4\sqrt{2}}\ln\left|\frac{x^2-\sqrt{2}x+1}{x^2+\sqrt{2}x+1}\right| + C$

(2) $\int\frac{dx}{\cos^4x}$

$= \int\frac{\sin^2x+\cos^2x}{\cos^4x}dx$

$= \int(\tan^2x\sec^2x+\sec^2x)dx$

$= \int\sec^2x(\tan^2x+1)dx$

$= \int(\tan^2x+1)\,d\tan x$

$= \frac{\tan^3x}{3}+\tan x+C$

(3) $\int\frac{\sin^3x}{\cos^4x}dx$

$= \int\tan^3x\sec x\,dx$

$= \int\tan^2x\,d\sec x$

$= \int(\sec^2x-1)\,d\sec x$

$= \frac{\sec^3x}{3}+\sec x+C$

(4) $\int\frac{dx}{(1+x)\sqrt{x^2+x+1}}$

$1-4\times1\times1<0$

令 $\sqrt{x^2+x+1}=x+t$

$x^2+x+1=x^2+2tx+t^2$

$\frac{1-t^2}{1-2t}=x$

$\int\frac{dt}{t^2-2t}$ 略

(5) $\int\frac{1}{x}\cdot\sqrt{\frac{1+x}{1-x}}dx$

令 $\sqrt{\frac{1+x}{1-x}}=t$

$\frac{1+x}{1-x}=t^2$

$1+x=t^2-t^2x$

$\frac{t^2-1}{t^2+1}=x$ $\quad \frac{t^2+1-1}{t^2+1}=x=1-\frac{1}{t^2+1}$

$\frac{dx}{dt}=\frac{2t}{(t^2+1)^2}$

$\int\frac{\cancel{t^2+1}}{t^2-1}\cdot t\cdot\frac{2t}{\cancel{(t^2+1)^2}\ t^2+1}dt$

$= \int\frac{2t^2}{(t^2-1)(t^2+1)}dt$

略

南開大學 作业纸

系别______ 班级______ 姓名______ 第 页

(1) $\int \frac{1}{2+\sin x}dx$

令 $t=\tan\frac{x}{2}$

∴ $x=2\arctan t$

$\frac{dx}{dt}=\frac{2}{1+t^2}$

$\sin x=\frac{2t}{1+t^2}$

$\int \frac{1}{2+\frac{2t}{t^2+1}}\cdot\frac{2}{t^2+1}dt$

$=\int \frac{\cancel{t^2+1}}{2t^2+2+2t}\cdot\frac{2}{\cancel{t^2+1}}dt$

$=\int \frac{1}{t^2+t+1}dt$

$=\int \frac{1}{(t+\frac{1}{2})^2+\frac{3}{4}}dt$

$=\int \frac{1}{(\frac{t+\frac{1}{2}}{\frac{\sqrt3}{2}})^2+1}dt$

$=\frac{4}{3}\int \frac{1}{(\frac{2}{\sqrt3}t+\frac{1}{\sqrt3})^2+1}dt$

$=\frac{4}{3}\cdot\frac{\sqrt3}{2}\int \frac{1}{(\frac{2}{\sqrt3}t+\frac{1}{\sqrt3})^2+1}d\frac{2}{\sqrt3}t$

$=\frac{2\sqrt3}{3}\arctan(\frac{2}{\sqrt3}t+\frac{1}{\sqrt3})+C$

$=\frac{2}{\sqrt3}\arctan\frac{2\tan\frac{x}{2}+1}{\sqrt3}+C$

(2) $\int\sqrt{1+\sin x}\,dx$

$=\int\sqrt{(\sin\frac{x}{2}+\cos\frac{x}{2})^2}\,dx$ 略

(3) $\int\frac{1}{1+\sin x+\cos x}dx$

$t=\tan\frac{x}{2}$ $\frac{dx}{dt}=\frac{2}{1+t^2}$

$\int\frac{1}{1+\frac{2t}{1+t^2}+\frac{1-t^2}{1+t^2}}\cdot\frac{2}{1+t^2}dt$

$=\int\frac{2}{t^2+1+2t+1-t^2}dt$

$=\int\frac{1}{t+1}dt$ 略

(4) $\int\frac{xe^x}{(1+x)^2}dx$

$=-\int xe^x\,d\frac{1}{x+1}$

$=-\frac{xe^x}{x+1}+\int\frac{1}{x+1}e^x(x+1)dx$

$=-\frac{xe^x}{x+1}+e^x+C$

(5) $\int(\arcsin x)^4dx$

令 $\arcsin x=t$ ∴ $x=\sin t$

$\int t^4\,d\sin t$

$=\sin t\cdot t^4-\int\sin t\cdot 3t^3dt$

$=\sin t\cdot t^4+3\int t^3\,d\cos t$

$=\sin t\,t^4+3t^3\cos t-3\int\sin t\,3t^2dt$

后略

(6) $\int\frac{dx}{1+\sqrt{1-2x-x^2}}$

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(1) $\int \frac{(1+x^2)\arcsin x}{x^2\sqrt{1-x^2}}dx$

$=\int \frac{(1+x^2)\arcsin x}{x^2} d\arcsin x$

$=\int \frac{\arcsin x}{x^2} d\arcsin x+\frac{(\arcsin x)^2}{2}+C$

令 $\arcsin x = t$

$\int \frac{t}{\sin^2 t}dt$

$=-t\cot t+\ln\sin t+C$

如果看不出来注意力是不是有点涣散

(2) $\int\sqrt{\tan x}\,dx$

令 $\tan x=t^2$

$x=\arctan t^2$

$\int t\, d\arctan t^2$

$\cancel{=t\arctan t^2-\int 2t\frac{2t}{1+t^4}dt}$

$\cancel{=t^2\arctan t^2-4\int\frac{t^2}{1+t^4}dt}$

接下来是显然的

$\int t\cdot\frac{2t}{1+t^4}dt$

$=\int\frac{2t^2}{1+t^4}dt$

(3) $\int\frac{1}{(2x^2+1)\sqrt{x^2+1}}dx$

令 $x=\tan t$

$\int\frac{\sec t\,dt}{(2\tan^2 t+1)}$

$=\int\frac{2\cos t\,dt}{1+\sin^2 t}$

$=2\int\frac{d\sin t}{1+\sin^2 t}$

$=2\arctan(\sin t)+C$

$=2\arctan(\sin\arctan x)+C$

(4) $\int\frac{\arcsin e^x}{e^x}dx$

令 $e^x=t$ $x=\ln t$ $\frac{dx}{dt}=\frac{1}{t}$

$\int\frac{\arcsin t}{t}\frac{dt}{t}$

$\cancel{=\int\arcsin t\, d\frac{1}{t}}$

$=-\frac{\arcsin t}{t}+\int\frac{1}{t}\cdot\frac{1}{\sqrt{1-t^2}}dt$

令 $\cancel{\sqrt{1-t^2}=m \quad t^2=m^2}$ $t^2=1-m^2$

$\cancel{\frac{dt}{dm}=\frac{-m}{\sqrt{1-m^2}}}$

$\cancel{\int(1-m^2)m}$

$\cancel{=-\frac{\arcsin t}{t}+\int\frac{1}{\sqrt{1-t^2}}d\ln t}$

$\cancel{=-\frac{\arcsin t}{t}+\sqrt{1-t^2}}$

$-e^{-x}\arcsin e^x+\int\frac{dx}{\sqrt{1-e^{2x}}}$ 令 $t=\sqrt{1-e^{2x}}$

$x=\frac{1}{2}\ln(1-t^2)$

$\int\frac{dt}{t^2-1}$ 略

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(1) $\int \frac{1}{(1+x^2)^3}dx$

1. 令 $I_n = \int \frac{1}{(1+x^2)^n}dx$

$I_1 = \frac{x}{1+x^2} + 2I_1 - 2I_2$

$I_2 = \frac{1}{2}\left(\frac{x}{1+x^2} + I_1\right)$

$\int \frac{1}{(1+x^2)^2}dx = \frac{x}{(1+x^2)^2} + 2\int x \cdot \frac{2x}{(1+x^2)^3}dx$

$= \frac{x}{(1+x^2)^2} + \int \frac{4x^2}{(1+x^2)^3}dx$

$= \frac{x}{(1+x^2)^2} + 4I_2 - 4I_3$

$I_3 = \frac{x}{4(1+x^2)^2} + \frac{3}{8}\frac{x}{1+x^2} + \frac{3}{8}\arctan x + C$

2. 令 $x = \tan\theta$ $\qquad \frac{dx}{d\theta} = \frac{1}{\cos^2\theta}$

$\int \frac{1}{(1+\tan^2\theta)^3}\frac{1}{\cos^2\theta}d\theta$

$= \int \cancel{\frac{1}{\cos^4\theta}d\theta}$

$= \int \cancel{\frac{\sin^2\theta+\cos^2\theta}{\cos^4\theta}}d\theta$

$= \int \cos^4\theta\, d\theta$

略 更简单

(2) $\int \ln\left(1+\sqrt{\frac{x+1}{x}}\right)dx$

令 $\sqrt{\frac{x+1}{x}} = t$

$\frac{x+1}{x} = t^2$

$1 + \frac{1}{x} = t^2 \qquad x = \frac{1}{1-t^2}$

$\frac{dx}{dt} = -\frac{-2t}{(1-t^2)^2} = \frac{2t}{(1-t^2)^2}$

$\int \ln(1+t)\frac{2t}{(1-t^2)^2}dt$

$= \int \ln(1+t)\, d\frac{1}{1-t^2}$

$= \frac{\ln(1+t)}{1-t^2} - \int \frac{1}{(1-t^2)}\cdot\frac{1}{1+t}dt$

$= x\ln\left(1+\sqrt{\frac{1+x}{x}}\right) + \frac{1}{4}\ln\left(1+\sqrt{\frac{x+1}{x}}\right)$

$-\frac{1}{2}\frac{1}{1+\sqrt{\frac{1+x}{x}}} - \frac{1}{4}\ln\left|\sqrt{\frac{x+1}{x}}-1\right| + C$

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